File Descriptors and Redirections

A file descriptor (`FD`) in Unix/Linux operating systems is a reference, maintained by the kernel, that allows the system to manage Input/Output (`I/O`) operations. It acts as a unique identifier for an open file, socket, or any other I/O resource. In Windows-based operating systems, this is known as a file handle. Essentially, the file descriptor is the system's way of keeping track of active `I/O` connections, such as reading from or writing to a file.

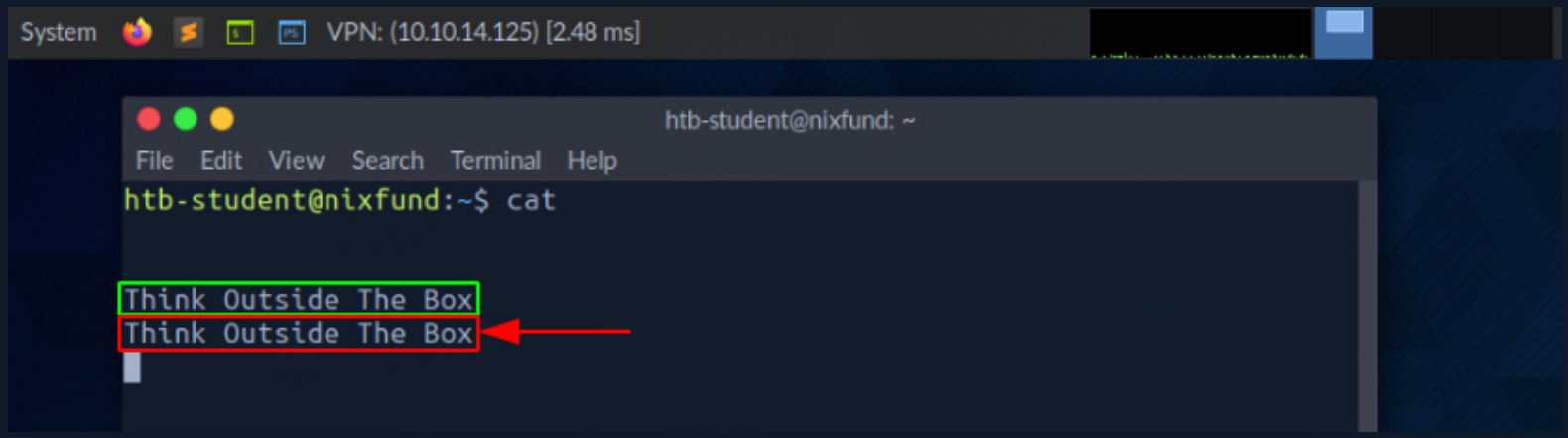
Think of it as a ticket number you get when checking in your coat at a coatroom. The ticket (file descriptor) represents your connection to your coat (file or resource), and whenever you need to retrieve your coat (perform I/O), you present the ticket to the attendant (operating system) who knows exactly where your coat is stored (which resource the file descriptor refers to). Without the ticket, you'd have no way of efficiently accessing your coat among the many others stored, just as without a file descriptor, the operating system wouldn't know which resource to interact with. You will soon see why file descriptors are so important and why understanding them is crucial as we dive into the upcoming examples.

By default, the first three file descriptors in Linux are:

1. Data Stream for Input
   - `STDIN - 0`
2. Data Stream for Output
   - `STDOUT - 1`
3. Data Stream for Output that relates to an error occurring.
   - `STDERR - 2`

STDIN and STDOUT

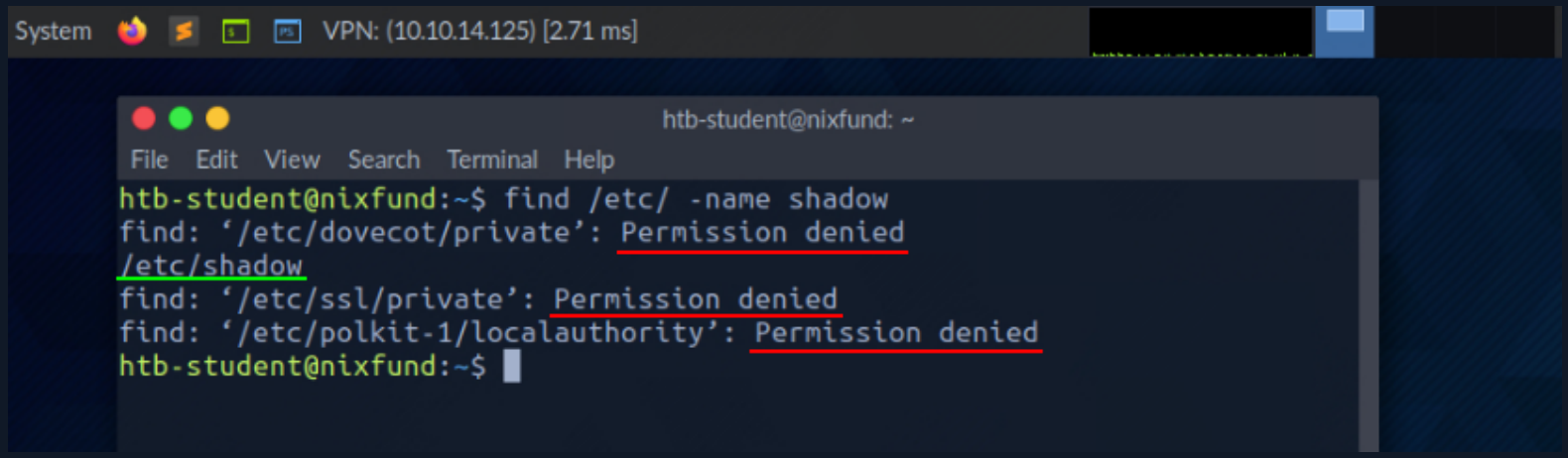
Let us see an example with `cat`. When running `cat`, we give the running program our standard input (`STDIN - FD 0`), marked `green`, wherein this case "SOME INPUT" is. As soon as we have confirmed our input with `[ENTER]`, it is returned to the terminal as standard output (`STDOUT - FD 1`), marked **red**.



STDOUT and STDERR

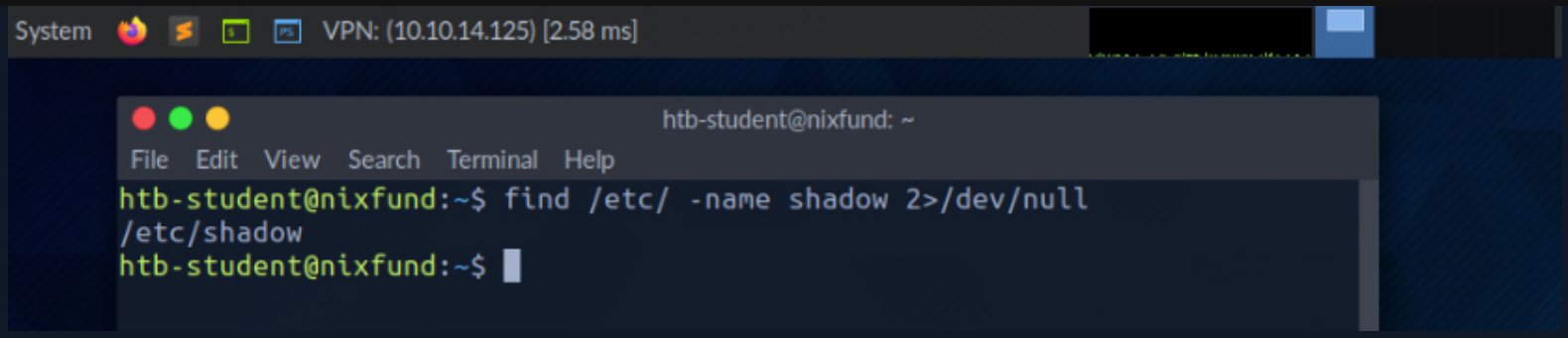
In the next example, by using the `find` command, we will see the standard output (`STDOUT - FD 1`) marked in `green` and standard error (`STDERR - FD 2`) marked in red.

```shellsession
athane@htb[/htb]$ find /etc/ -name shadow
```



In this case, the error is marked and displayed with " `Permission denied` ". We can check this by redirecting the file descriptor for the errors (`FD 2 - STDERR`) to " `/dev/null` ." This way, we redirect the resulting errors to the "null device," which discards all data.

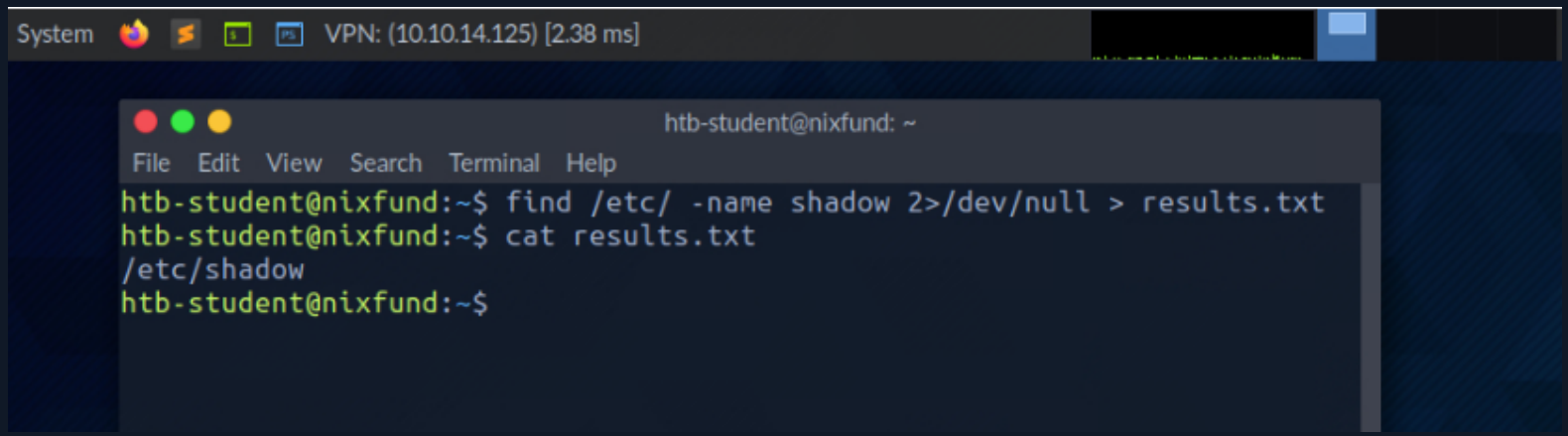
```shellsession
athane@htb[/htb]$ find /etc/ -name shadow 2>/dev/null
```



Redirect STDOUT to a File

Now we can see that all errors (`STDERR`) previously presented with " `Permission denied` " are no longer displayed. The only result we see now is the standard output (`STDOUT`), which we can also redirect to a file with the name `results.txt` that will only contain standard output without the standard errors.

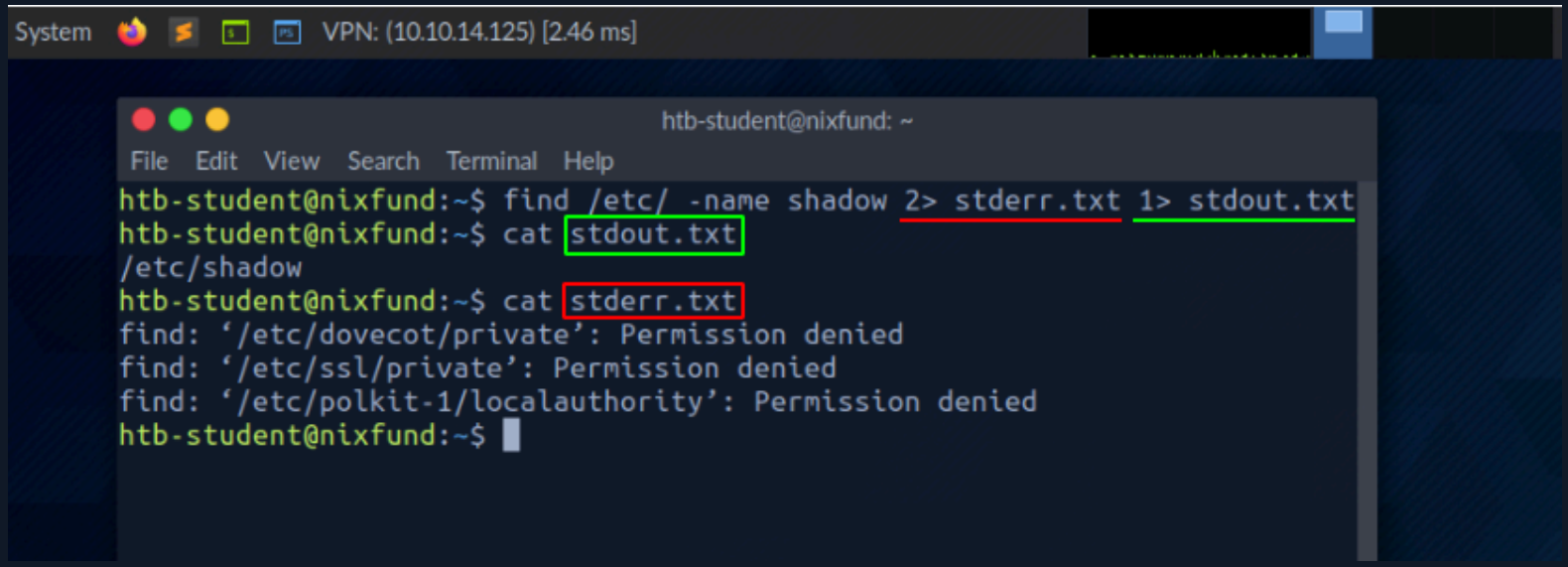
```shellsession
athane@htb[/htb]$ find /etc/ -name shadow 2>/dev/null > results.txt
```



Redirect STDOUT and STDERR to Separate Files

We should have noticed that we did not use a number before the greater-than sign (`>`) in the last example. That is because we redirected all the standard errors to the " `null device` " before, and the only output we get is the standard output (`FD 1 - STDOUT`). To make this more precise, we will redirect standard error (`FD 2 - STDERR`) and standard output (`FD 1 - STDOUT`) to different files.

```shellsession
athane@htb[/htb]$ find /etc/ -name shadow 2> stderr.txt 1> stdout.txt
```

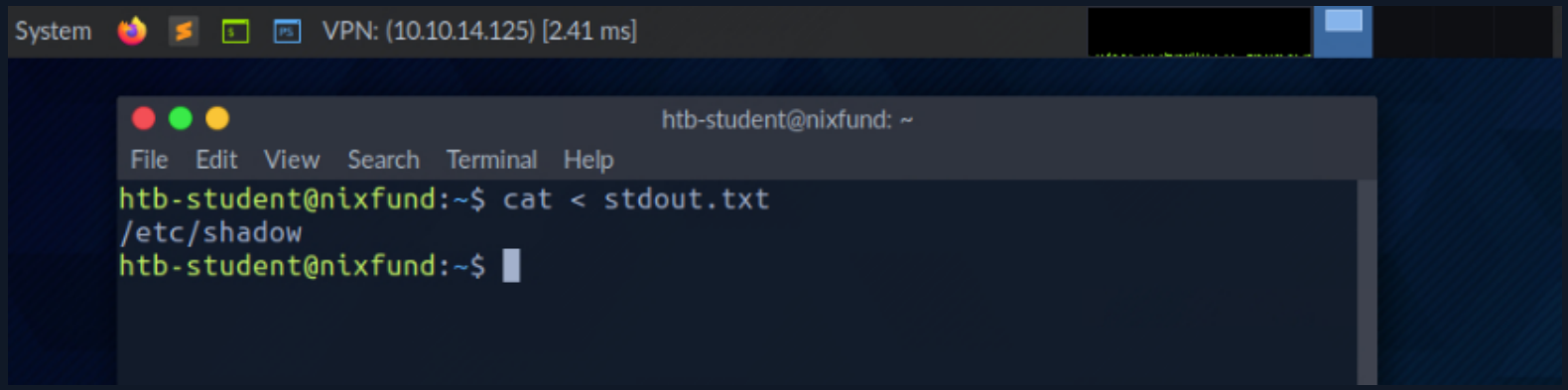


Redirect STDIN

As we have already seen, in combination with the file descriptors, we can redirect errors and output with greater-than character (`>`). This also works with the lower-than sign (`<`). However,

the lower-than sign serves as standard input (`FD 0 - STDIN`). These characters can be seen as " `direction` " in the form of an arrow that tells us " `from where` " and " `where to` " the data should be redirected. We use the `cat` command to use the contents of the file " `stdout.txt` " as `STDIN` .

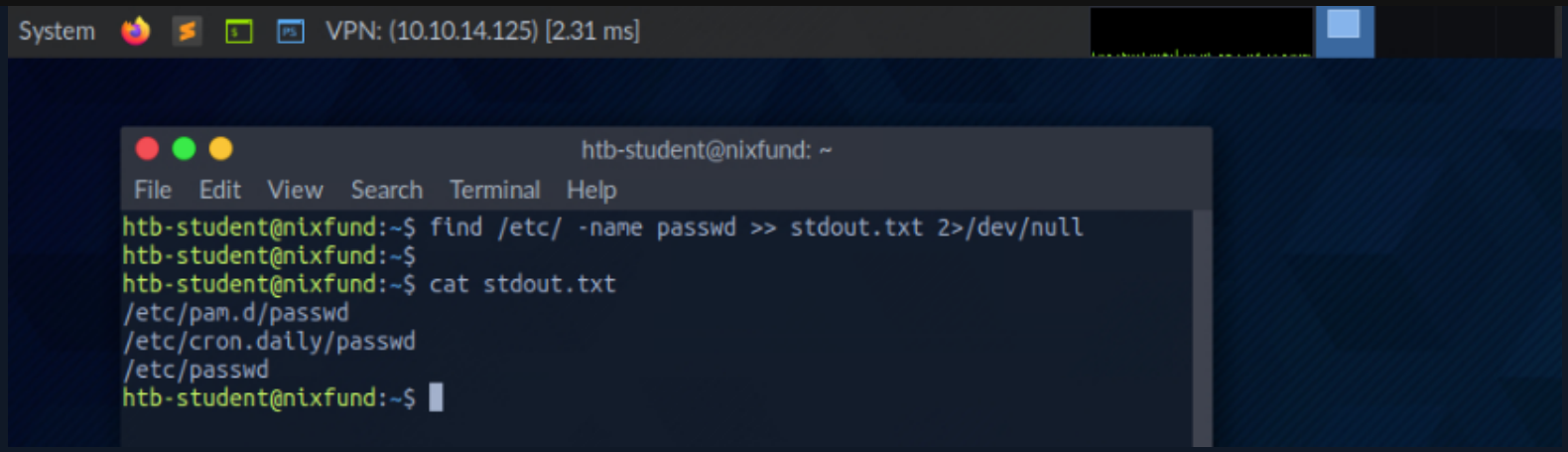
```shellsession
athane@htb[/htb]$ cat < stdout.txt
```



Redirect STDOUT and Append to a File

When we use the greater-than sign (`>`) to redirect our `STDOUT` , a new file is automatically created if it does not already exist. If this file exists, it will be overwritten without asking for confirmation. If we want to append `STDOUT` to our existing file, we can use the double greater-than sign (`>>`).

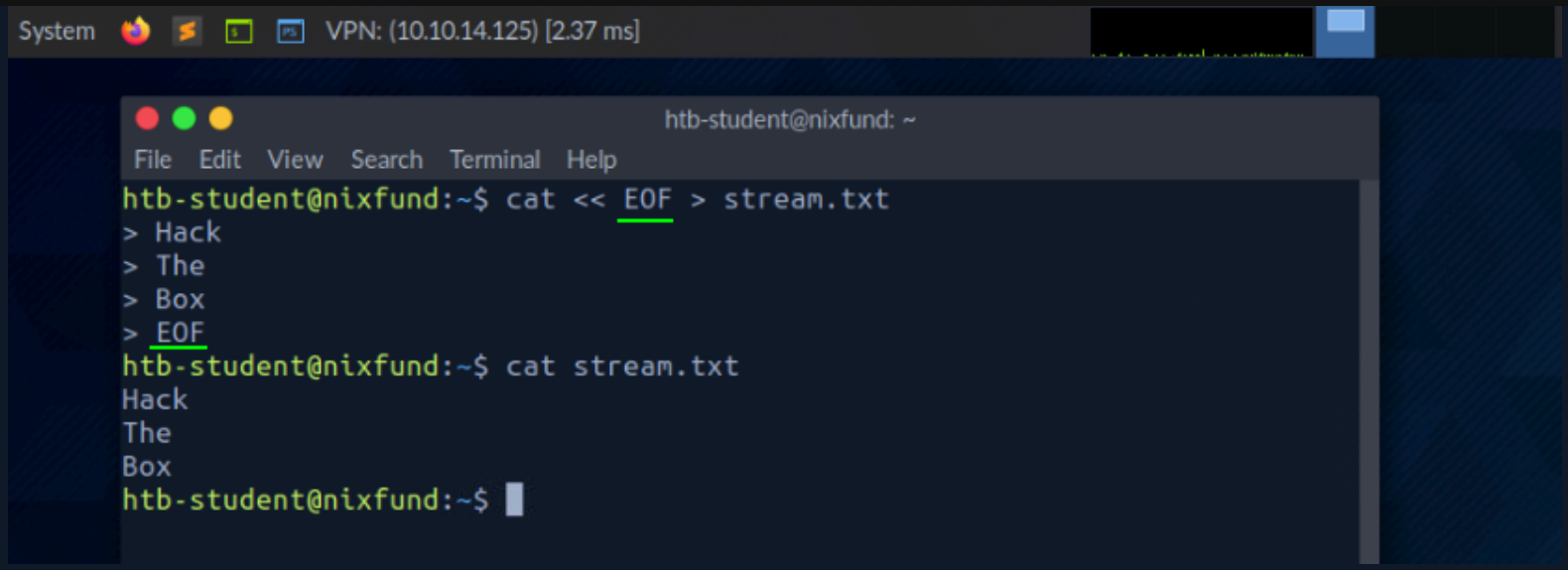
```shellsession
athane@htb[/htb]$ find /etc/ -name passwd >> stdout.txt 2>/dev/null
```



Redirect STDIN Stream to a File

We can also use the double lower-than characters (`<<`) to add our standard input through a stream. We can use the so-called `End-Of-File` (`EOF`) function of a Linux system file, which defines the input's end. In the next example, we will use the `cat` command to read our streaming input through the stream and direct it to a file called " `stream.txt` ."

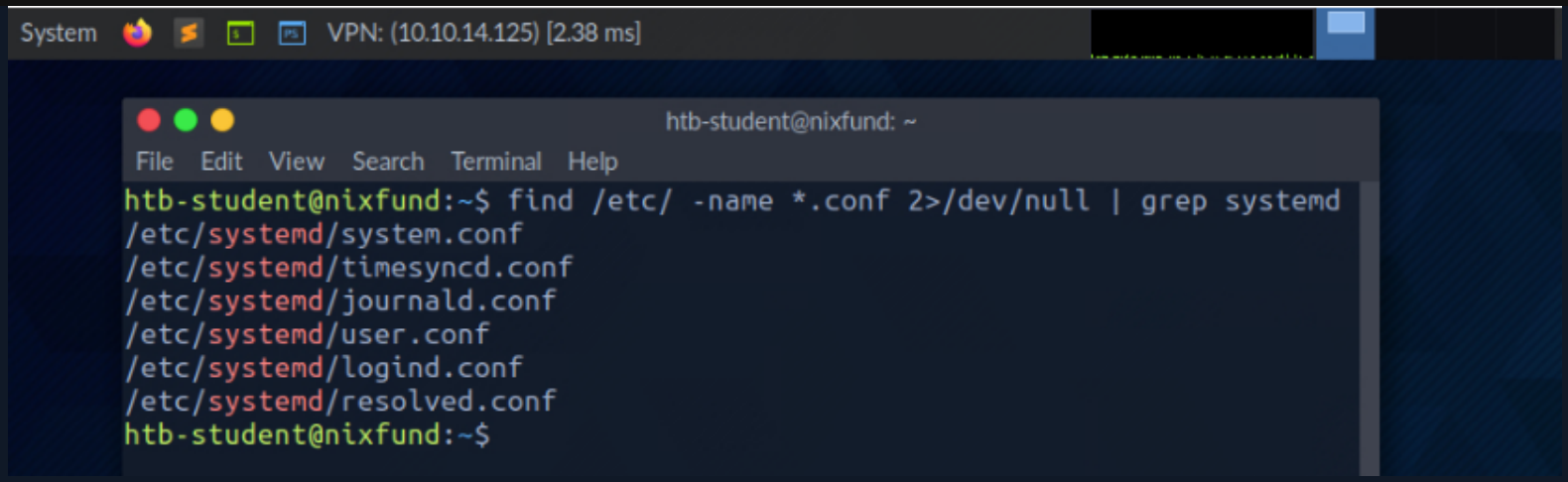
```shellsession
athane@htb[/htb]$ cat << EOF > stream.txt
```



Pipes

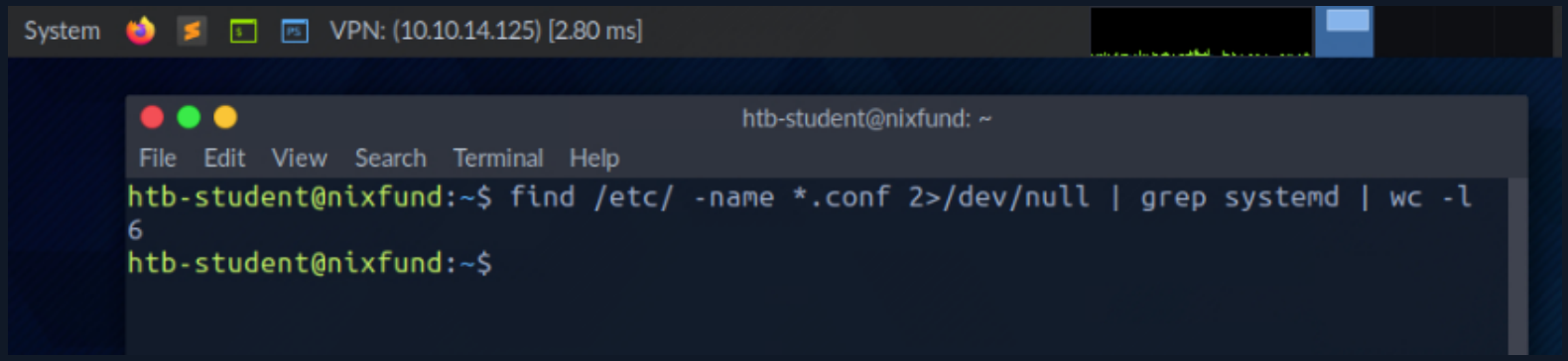
Another way to redirect `STDOUT` is to use pipes (`|`). These are useful when we want to use the `STDOUT` from one program to be processed by another. One of the most commonly used tools is `grep` , which we will use in the next example. Grep is used to filter `STDOUT` according to the pattern we define. In the next example, we use the `find` command to search for all files in the " `/etc/` " directory with a " `.conf` " extension. Any errors are redirected to the " `null device` " (`/dev/null`). Using `grep` , we filter out the results and specify that only the lines containing the pattern " `systemd` " should be displayed.

```shellsession
athane@htb[/htb]$ find /etc/ -name *.conf 2>/dev/null | grep systemd
```



The redirections work, not only once. We can use the obtained results to redirect them to another program. For the next example, we will use the tool called `wc`, which should count the total number of obtained results.

```shellsession
athane@htb[/htb]$ find /etc/ -name *.conf 2>/dev/null | grep systemd | wc -l
```



Now that we have a fundamental understanding of file descriptors, redirections, and pipes, we can structure our commands more efficiently to extract the exact information we need. This

knowledge allows us to manipulate how input and output flows between files, processes, and the system, enabling us to handle data more effectively. By leveraging these tools, we can streamline tasks, avoid unnecessary steps, and work with files and system resources in a much more organized and efficient manner, ultimately enhancing our productivity and precision in managing operations.

Connect to HTB

Target(s)
Spawn the target system to get IPs and answer questions

Spawn the target system

Enable step-by-step solutions PRO

Question 1 +1 +20

How many files exist on the system that have the ".log" file extension?

SSH to with user "**htb-student**" and password "**HTB_@cademy_stdnt!**"

Write your answer

Submit

Question 2

+20

How many total packages are installed on the target system?

Write your answer

Submit